

FORM TP 2015205



TEST CODE **02125020**

MAY/JUNE 2015

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

GEOGRAPHY

**POPULATION GEOGRAPHY, GEOMORPHIC
PROCESSES AND HAZARDS**

UNIT 1 – Paper 02

3 hours

15 MAY 2015 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SEVEN questions. Answer FOUR questions as indicated below.
2. Section A consists of ONE question testing practical skills. This is compulsory.
3. Section B consists of SIX questions. Answer THREE questions, ONE from EACH module.
4. Write your answers in the answer booklet provided.
5. A map extract and a grid are provided for Question 1.
6. All diagrams must be well labelled.
7. You may use a silent, non-programmable calculator to answer questions.
8. You may use geometrical instruments.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Copyright © 2013 Caribbean Examinations Council
All rights reserved.

02125020/CAPE 2015



SECTION A

Answer this compulsory question.

1. (a) Study the map extract of Kingston, Jamaica (**provided as an insert**), which is on a scale of 1:50 000, and use the information to answer the following questions.
- (i) a) Describe the distribution of TWO types of settlement patterns evident on the map extract. [6 marks]
 - b) Explain how the TWO types of settlement patterns you described in (a) (i) a) are influenced by relief. [6 marks]
 - c) Examine the settlement of Bull Bay and outline ONE factor that has influenced its location. [3 marks]
 - (ii) On the grid in Figure 1 (**provided as an insert**), draw a labelled sketch map to show the coastline as well as the main relief and drainage features of the area bounded by Eastings 75 and 85, and Northings 42 and 52. [9 marks]
 - (iii) Suggest an explanation for the formation of
 - a) ONE coastal landform in the area covered by the map extract [3 marks]
 - b) ONE feature of the channel of the Yallahs River. [3 marks]
 - (iv) Using appropriate map evidence, suggest how THREE natural and/or technological hazards are **LIKELY** to affect the area covered by the map extract. [9 marks]
 - (b) Figure 2 (**page 3**) shows a map of the epicentres of major earthquakes in Saint Lucia since 1982.

Using map evidence **ONLY**, suggest **THREE** reasons why Castries is a larger settlement than Soufriere. [6 marks]

Total 45 marks

GO ON TO THE NEXT PAGE

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

GEOGRAPHY

UNIT 1 – Paper 02

Insert for Question 1. (a) (ii)

Centre Number _____

Candidate's Number _____

Title: _____

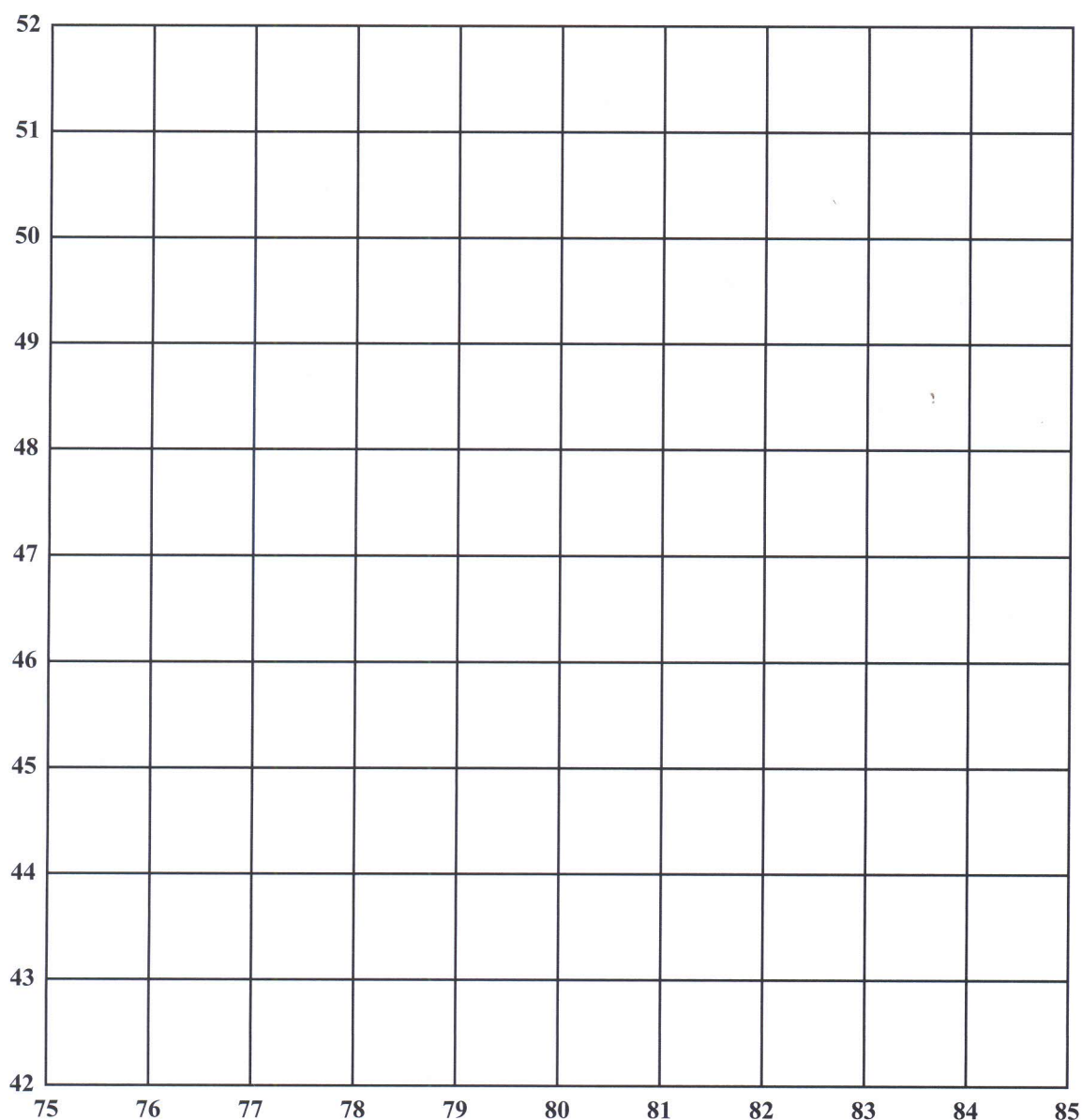


Figure 1. Sketch map showing relief and drainage

TO BE ATTACHED TO YOUR ANSWER BOOKLET

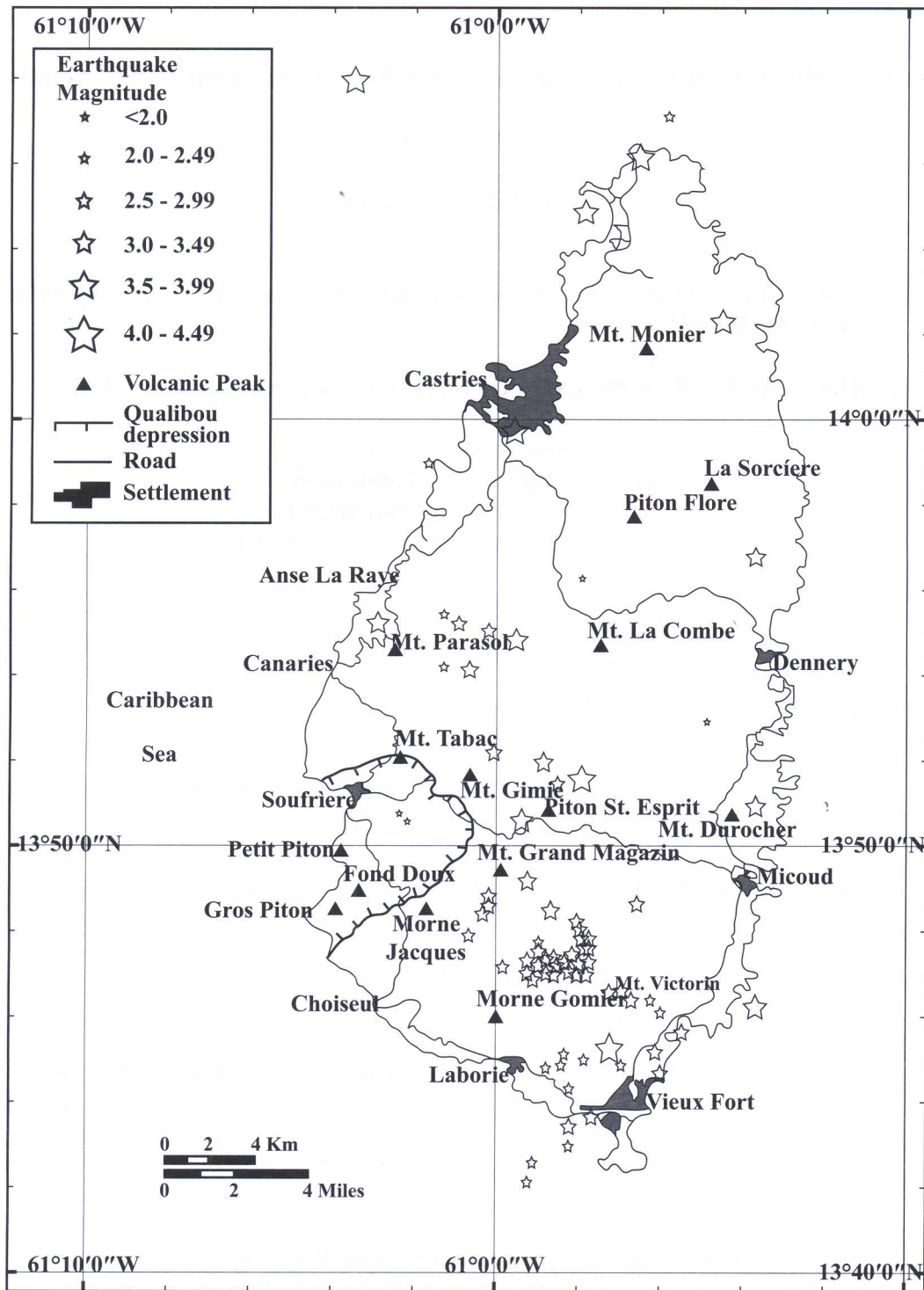


Figure 2. Map showing epicentres of major earthquakes in Saint Lucia since 1982

Source: *Volcanic Hazard Atlas of the Lesser Antilles*, Seismic Research Unit, The University of the West Indies, 2005, p. 227.

GO ON TO THE NEXT PAGE

SECTION B

Answer **THREE** questions from this section, **ONE** question from **EACH** module.

MODULE 1

Answer **ONE** question.

2. (a) Study Table 1 which shows age groups and population for Country X, and answer the questions that follow.

TABLE 1: AGE GROUPS AND POPULATION DATA FOR COUNTRY X

Age Group	Population (thousands)
0–15	11 345
16–64	38 498
65 and over	9 271

- (i) Using the data presented in Table 1, calculate the dependency ratio for the population of Country X. **[3 marks]**
- (ii) What type of country would be indicated by the dependency ratio calculated in (a) (i)? **[1 mark]**
- (b) (i) Identify the technique that may be used to depict population distribution on a map. **[1 mark]**
- (ii) Describe **THREE** factors to be considered when preparing a map using the technique in (b) (i). **[6 marks]**
- (iii) Name **ONE** other technique that may be used to depict population distribution. **[1 mark]**
- (c) Write an essay to discuss **THREE** differences between the population structure of a more developed country (MDC) and a less developed country (LDC). **[14 marks]**
- (d) Explain how the shape of population pyramids of LDCs or MDCs may change over time. **[4 marks]**

Total 30 marks

GO ON TO THE NEXT PAGE

3. (a) Study Figure 3 which shows the relationship between population growth and food supply, and answer the questions that follow.

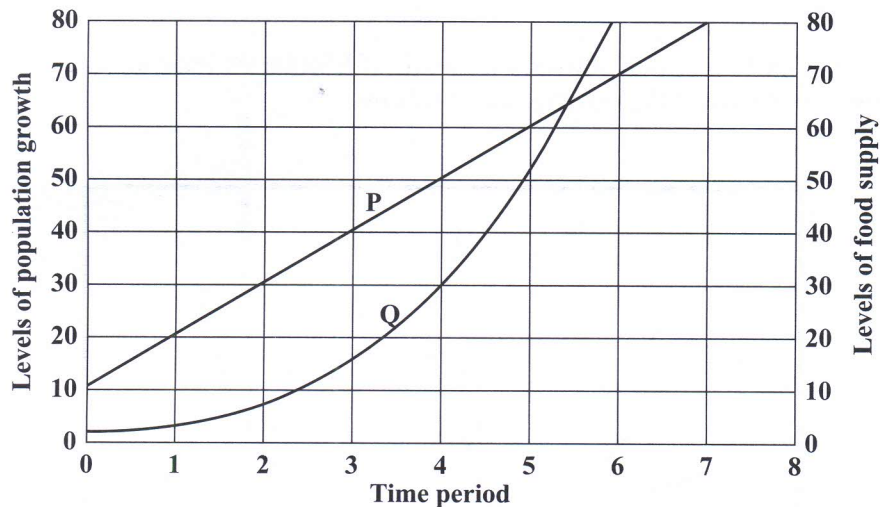


Figure 3. Relationship between population growth and food supply according to Malthus

- (i) State an appropriate label for EACH of the lines, P and Q, shown in the figure. [2 marks]
- (ii) Compare the trends shown by the two lines in (a) (i). [4 marks]
- (b) Outline FOUR factors that may contribute to a declining population in rural areas. [8 marks]
- (c) Explain the meaning of the following statement.
“Most modern settlements, even those in rural areas, are multi-functional.” [4 marks]
- (d) Write an essay to explain why the carrying capacity of an area seldom remains static. Include THREE reasons in your response. [12 marks]

Total 30 marks

GO ON TO THE NEXT PAGE

MODULE 2

Answer ONE question.

4. (a) Study Figure 4 which shows a network of tributaries entering the main river in a river basin, and answer the questions that follow.

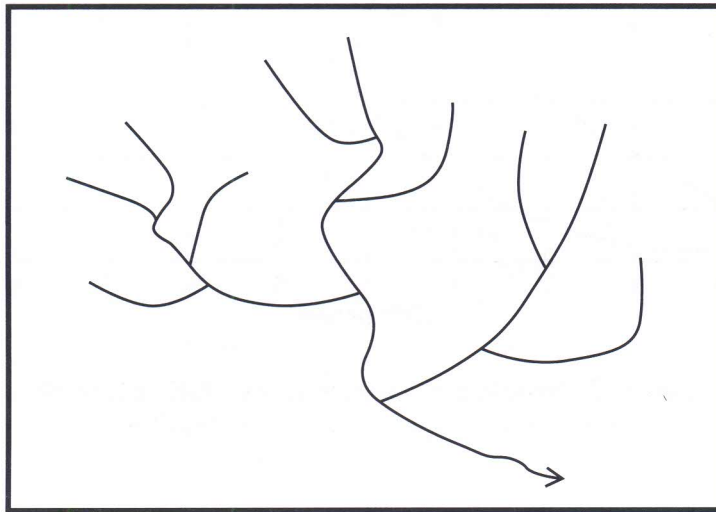


Figure 4. Network of streams in a river basin

For the river basin shown in Figure 4, determine

- (i) the number of first order streams [1 mark]
 - (ii) the HIGHEST stream order [1 mark]
 - (iii) the bifurcation ratio. [2 marks]
- (b) (i) State the water balance equation. [2 marks]
- (ii) Outline how the drainage density of a river basin is calculated. [2 marks]
- (c) Explain how rock type and structure influence the development of TWO drainage patterns. [8 marks]
- (d) Discuss how the type and rate of physical weathering varies with THREE types of climate. [14 marks]

Total 30 marks

GO ON TO THE NEXT PAGE

5. (a) Study Figure 5 which shows a section of a coastline where coastal structures have been erected, and answer the questions that follow.

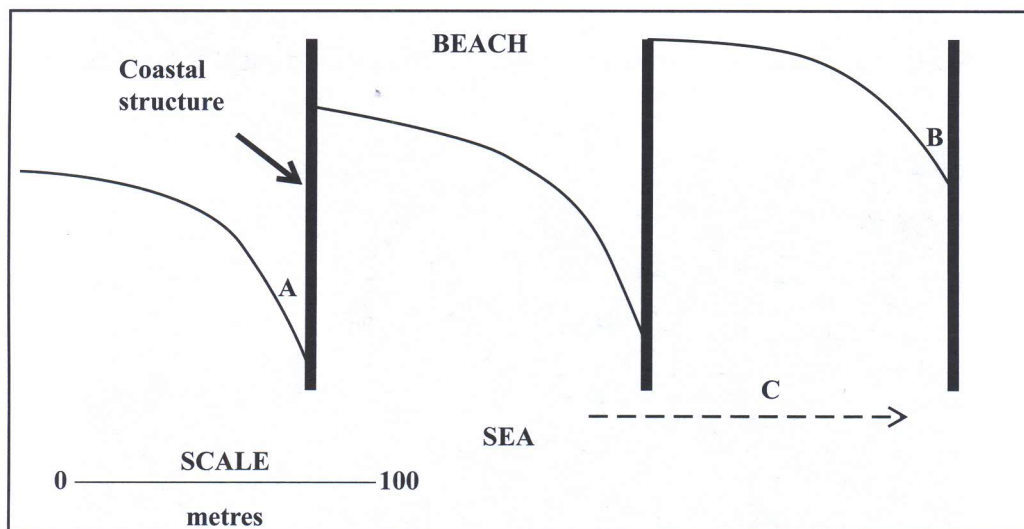


Figure 5. Section of a coastline

- (i) What is the length of the coastal structure at A? [1 mark]
- (ii) Identify the coastal process occurring at B. [1 mark]
- (iii) What process is represented by Arrow C? [1 mark]
- (iv) What is the name of the man-made coastal structure? [1 mark]
- (b) (i) State TWO causes of local sea level change and TWO causes of global sea level change. [4 marks]
- (ii) Describe TWO coastal features that result from a rise in sea level? [4 marks]
- (c) Explain the formation of any TWO surface features resulting from chemical weathering in limestone rocks. [6 marks]
- (d) Explain how FOUR different types of human activity threaten coral reefs in the Caribbean. [12 marks]

Total 30 marks

MODULE 3

Answer ONE question.

6. (a) Study Figure 6 which shows a response to a hazard, and answer the questions that follow.



Figure 6. Response to a hazard

Source: <http://www.aucklandcivildefence.org.nz/get-ready>

- (i) Identify the TYPE of hazard shown in the figure. **[1 mark]**
- (ii) Suggest a possible cause of the hazard. **[1 mark]**
- (iii) List TWO impacts of the hazard shown in the figure. **[2 marks]**
- (b) Outline FOUR measures that can be used at the national level to reduce the negative effects of hazards. **[8 marks]**
- (c) “Earthquake prediction is an immature science as it cannot predict the location, date and magnitude of an event with any reliability.”

Write an essay to discuss the statement above with respect to THREE short-term methods of earthquake prediction. **[18 marks]**

Total 30 marks

7. (a) Study Figure 7 which shows three fault blocks labelled A, B and C, and answer the questions that follow.

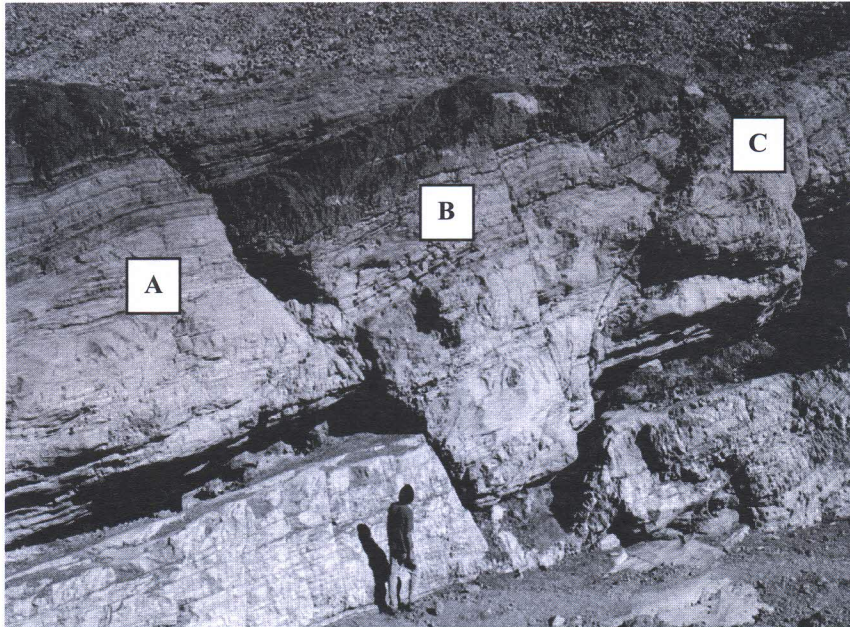


Figure 7. Fault blocks

Source: <http://marlimillerphoto.com/SrF.html>

- (i) What type of fault is shown in the figure? [1 mark]
 - (ii) State ONE reason for the answer given in (a) (i). [1 mark]
 - (iii) What is the name given to the feature lying between the two faults? [1 mark]
 - (iv) Identify the force which produced the fault blocks shown in the figure. [1 mark]
- (b) Describe the processes which occur when two continental plates converge. [8 marks]
- (c) Explain how the magnitude of a flood is affected by
- (i) FOUR characteristics of the drainage basin [12 marks]
 - (ii) TWO drainage patterns. [6 marks]

Total 30 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.